

Sale Price, Sales Tax, and Finding the Original Amount

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Standard: CCSS.MATH.CONTENT.7.RP.A.3 — Use proportional relationships to solve multistep ratio and percent problems including discounts, sales tax, and finding the whole given a part and the percent.

Quick Reference — Forward and Reverse

- **Forward (find the new price):** $\text{Sale Price} = \text{Original} \times (1 - \text{rate})$; $\text{Total Cost} = \text{Original} \times (1 + \text{rate})$
- **Reverse (find the original):** $\text{Original} = \text{Sale Price} \div (1 - \text{rate})$; $\text{Original} = \text{Total Cost} \div (1 + \text{rate})$
- **Find the rate:** $\text{rate} = (\text{change} \div \text{original}) \times 100\%$
- **Multi-step:** Apply the discount FIRST, then add sales tax to the discounted price.

Part A — Find the Sale Price

Directions: Find the sale price after the discount.

1. A jacket originally costs \$90 and is on sale for 30% off. Find the sale price.

Sale price: \$ _____

2. A pair of boots that costs \$120 has a 35% off sign. Find the sale price.

Sale price: \$ _____

3. A laptop priced at \$800 is on sale for 15% off. Find the sale price.

Sale price: \$ _____

4. A bicycle that normally costs \$250 is marked down 60%. Find the sale price.

Sale price: \$ _____

Part B — Find the Total Cost with Sales Tax

Directions: *Find the total cost after sales tax is added.*

5. A board game costs \$40, and there is a 5% sales tax. Find the total cost.

Total cost: \$ _____

6. A pair of shoes priced at \$75 has an 8% sales tax. Find the total cost.

Total cost: \$ _____

7. A water bottle costs \$30 and has a 10% sales tax. Find the total cost.

Total cost: \$ _____

8. A camera priced at \$200 has a 7% sales tax. Find the total cost.

Total cost: \$ _____

Part C — Find the Original Price from the Sale Price

Directions: *The discounted price is given. Work backward to find the original price before the discount. Use $\text{Original} = \text{Sale Price} \div (1 - \text{rate})$.*

9. After 25% off, a shirt costs \$60. What was the original price?

Original price: \$ _____

10. After 20% off, a watch sells for \$40. What was the original price?

Original price: \$ _____

11. After 50% off, a backpack costs \$35. What was the original price?

Original price: \$ _____

12. A book is on sale for 10% off and now costs \$27. What was the original price?

Original price: \$ _____

Part D — Find the Original Price from the Total Cost

Directions: *The total cost (with sales tax already added) is given. Work backward to find the price before tax. Use $\text{Original} = \text{Total Cost} \div (1 + \text{rate})$.*

13. Including a 5% sales tax, a meal costs \$63. What was the price before tax?

Original price: \$ _____

14. Including a 10% sales tax, a hat costs \$44. What was the price before tax?

Original price: \$ _____

15. Including a 4% sales tax, a notebook costs \$26. What was the price before tax?

Original price: \$ _____

16. Including a 20% sales tax, a model car costs \$48. What was the price before tax?

Original price: \$ _____

Part E — Find the Rate

Directions: Use $\text{rate} = (\text{change} \div \text{original}) \times 100\%$ to find the missing percent.

17. A toy was originally 80 and is now on sale for 60. What is the discount rate (% off)?

Discount rate: _____ %

18. A pen was originally priced at 20, and the total cost with sales tax was 21. What is the sales tax rate?

Tax rate: _____ %

Part F — Multi-Step (Discount, Then Tax)

Directions: Apply the discount first, then add sales tax to the discounted price.

19. A sweater is originally priced at \$80 and is on sale for 25% off. After the discount, a 10% sales tax is applied. Find the final total cost.

Final total: \$ _____

20. A skateboard regularly costs \$100. The store offers 40% off, then adds a 5% sales tax to the discounted price. Find the final total cost.

Final total: \$ _____